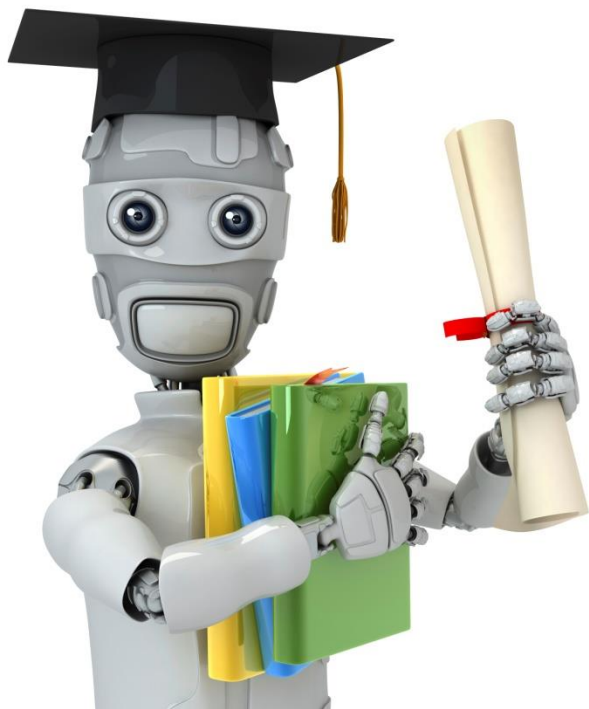




Machine Learning

Logistic Regression

Classification

Classification

Email: Spam / Not Spam?

Online Transactions: Fraudulent (Yes / No)?

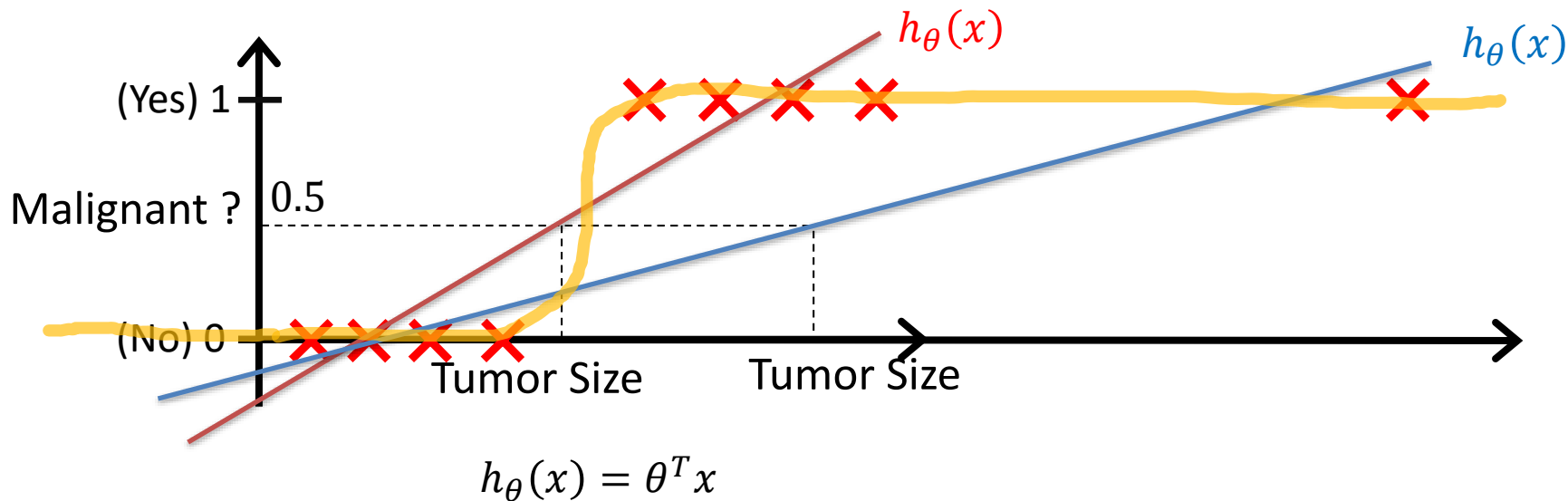
Tumor: Malignant / Benign ?

$$y \in \{0, 1\}$$


0: “Negative Class” (e.g., benign tumor)

1: “Positive Class” (e.g., malignant tumor)

二分类



Threshold classifier output $h_{\theta}(x)$ at 0.5:

If $h_{\theta}(x) \geq 0.5$, predict "y = 1"

If $h_{\theta}(x) < 0.5$, predict "y = 0"

Classification: $y = 0$ or 1

• $h_{\theta}(x)$ can be > 1 or < 0

线性回归

★ Logistic Regression: $0 \leq h_{\theta}(x) \leq 1$

映射

1. 逻辑回归是监督学习算法吗?

A. ☒ 是

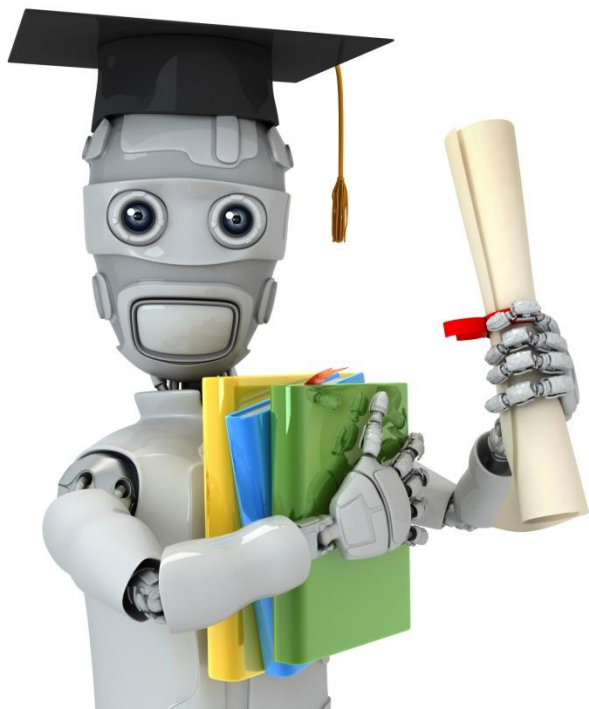
B. 不是

2. 逻辑回归主要用来做回归吗?

A. 是

B. ☒ 不是

分类



Machine Learning

Logistic Regression

Hypothesis Representation

Logistic Regression Model

Want $0 \leq h_{\theta}(x) \leq 1$

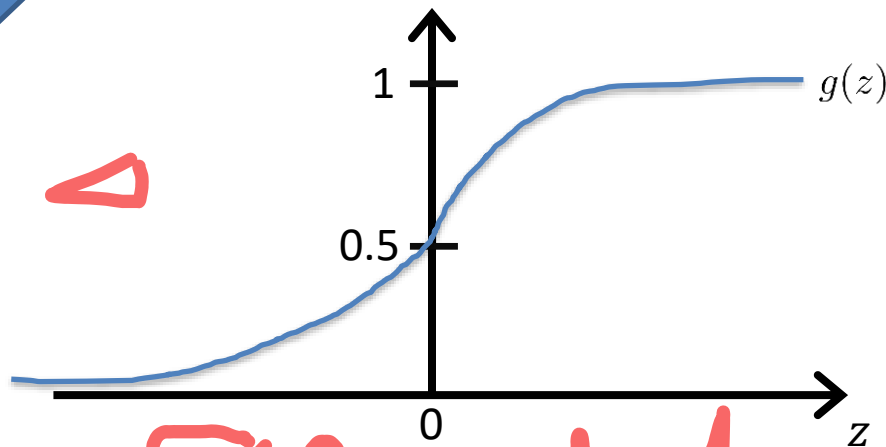
$y = \theta^T x$

$$h_{\theta}(x) = g(\theta^T x)$$

变量替换

$$g(z) = \frac{1}{1 + e^{-z}}$$

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$



Sigmoid

Sigmoid function

Logistic function

Interpretation of Hypothesis Output

$h_{\theta}(x)$ = estimated probability that $y = 1$ on input x

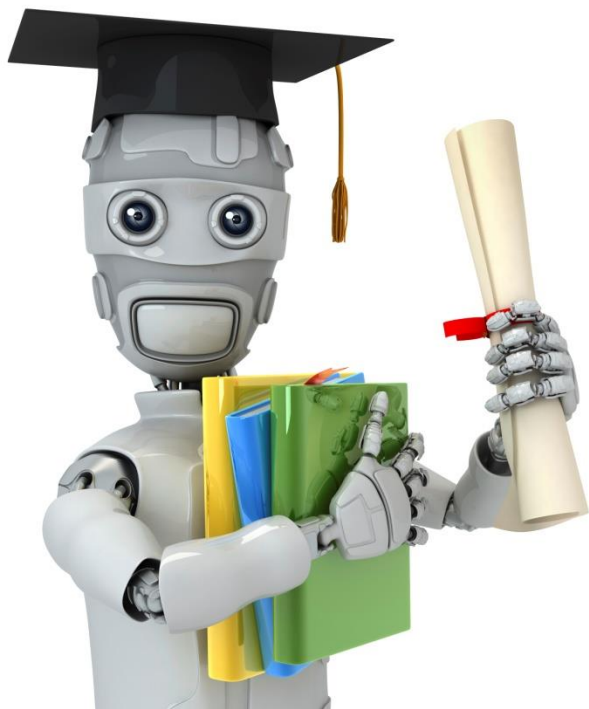
Example: If $x = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ \text{tumorSize} \end{bmatrix}$

$$h_{\theta}(x) = 0.7$$

Tell patient that 70% chance of tumor being malignant

“probability that $y = 1$, given x ,
parameterized by θ ”

$$\left\{ \begin{array}{l} P(y = 0|x; \theta) + P(y = 1|x; \theta) = 1 \\ P(y = 0|x; \theta) = 1 - P(y = 1|x; \theta) \end{array} \right.$$



Machine Learning

Logistic Regression

Decision boundary

决策边界

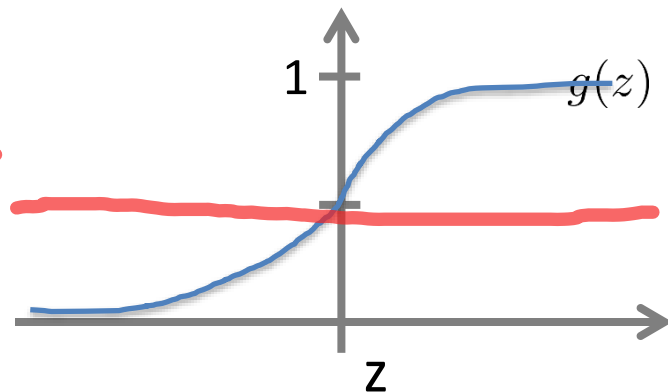
$0 = \theta^T x$ 的图形

Logistic regression

$$h_{\theta}(x) = g(\theta^T x)$$

$$g(z) = \frac{1}{1+e^{-z}}$$

$$y=0.5$$



Suppose predict “ $y = 1$ ” if $h_{\theta}(x) \geq 0.5$

$$z \geq 0, \quad g(z) \geq 0.5$$

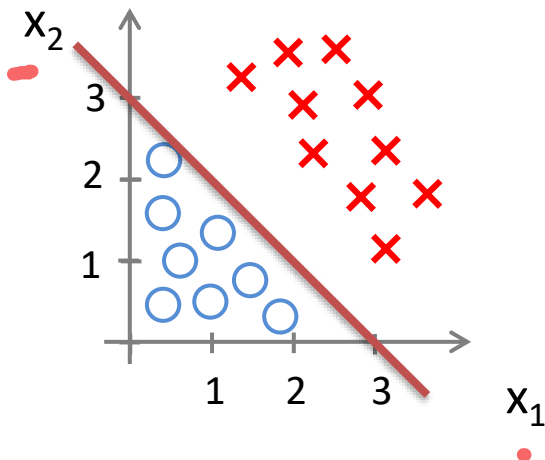
$$h_{\theta}(x) = g(\theta^T x) \geq 0.5 \quad \underline{\theta^T x \geq 0}$$

predict “ $y = 0$ ” if $h_{\theta}(x) < 0.5$

$$z < 0, \quad g(z) < 0.5$$

$$h_{\theta}(x) = g(\theta^T x) < 0.5 \quad \underline{\theta^T x < 0}$$

Decision Boundary



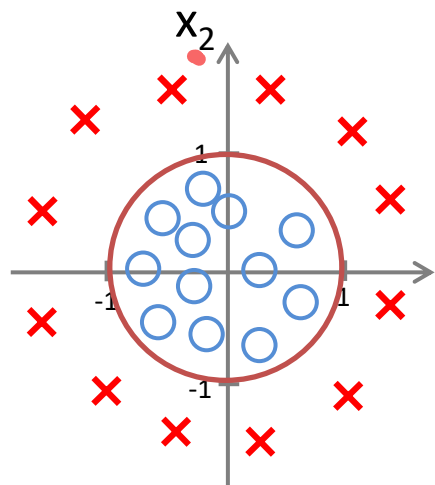
$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

$$(\theta_0, \theta_1, \theta_2) \cdot \begin{pmatrix} 1 \\ x_1 \\ x_2 \end{pmatrix} = y$$

Predict " $y = 1$ " if $-3 + x_1 + x_2 \geq 0$

Non-linear decision boundaries

入通项符号

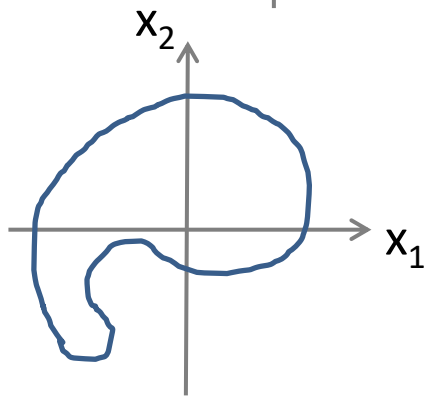


$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2$$

$$+ \theta_3 x_1^2 + \theta_4 x_2^2) \quad x_1, x_2$$

二次项

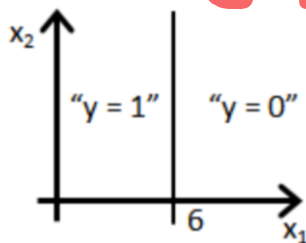
Predict “ $y = 1$ ” if $-1 + x_1^2 + x_2^2 \geq 0$



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^2 x_2^2 + \theta_6 x_1^3 x_2 + \dots)$$

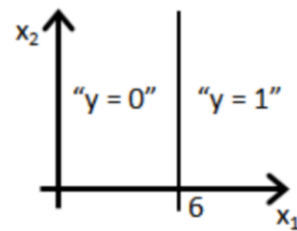
Suppose you train a logistic classifier $h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$. Suppose $\theta_0 = -6, \theta_1 = 1, \theta_2 = 0$. Which of the following figures represents the decision boundary found by your classifier?

Figure:



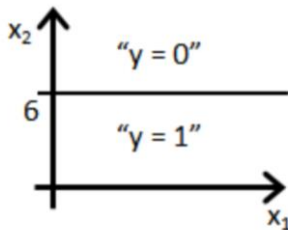
A.

Figure:



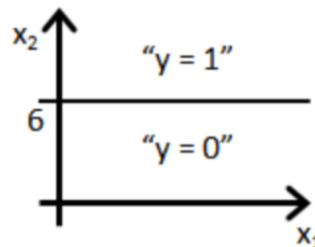
B.

Figure:



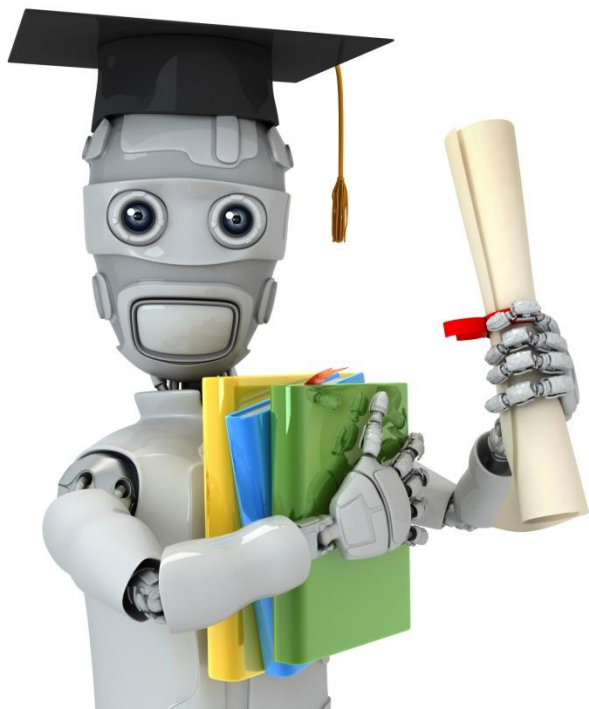
C.

Figure:



D.

B



Machine Learning

Logistic Regression

Cost function

Training set: $\{(\underline{x^{(1)}}), \underline{y^{(1)}}), (\underline{x^{(2)}}), \underline{y^{(2)}}), \dots, (\underline{x^{(m)}}), \underline{y^{(m)}})\}$

m examples

$$x \in \begin{bmatrix} x_0 \\ x_1 \\ \dots \\ x_n \end{bmatrix}$$

$$\underline{x_0 = 1, y \in \{0, 1\}}$$

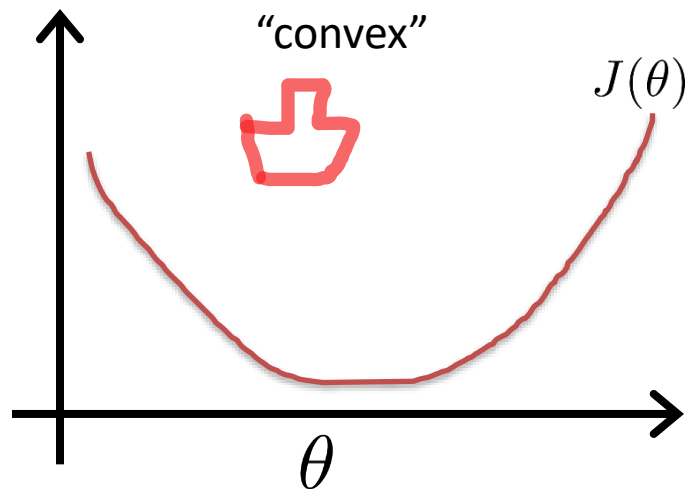
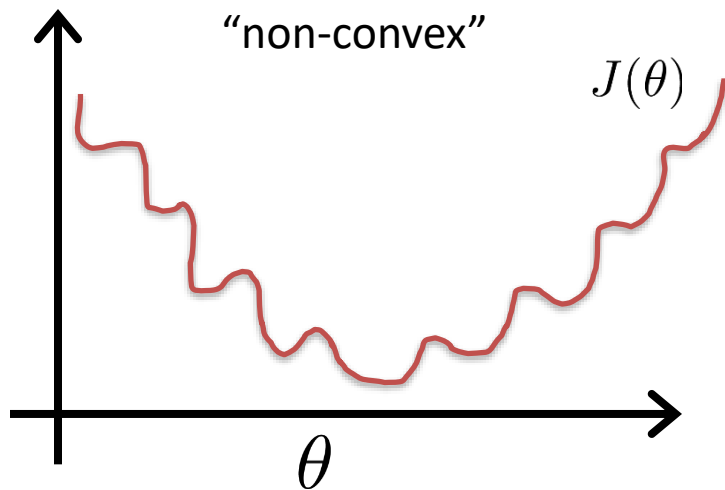
$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

How to choose parameters θ ?

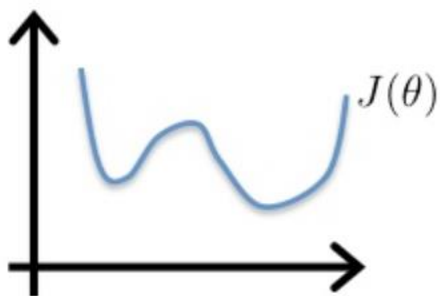
Cost function

Linear regression: $J(\theta) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} (h_{\theta}(x^{(i)}) - y^{(i)})^2$

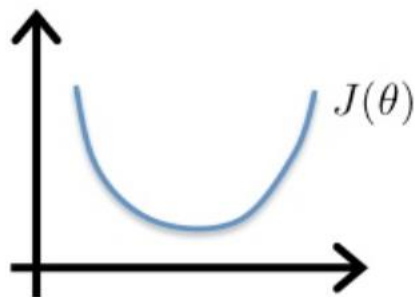
$$\text{Cost}(h_{\theta}(x^{(i)}), y^{(i)}) = \frac{1}{2} (h_{\theta}(x^{(i)}) - y^{(i)})^2$$



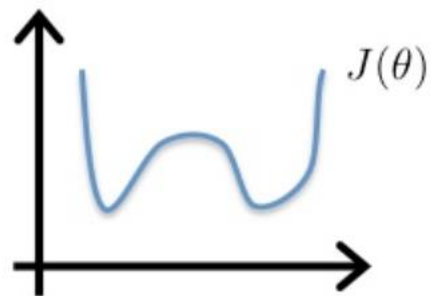
Consider minimizing a cost function $J(\theta)$, Which one of these functions is convex?



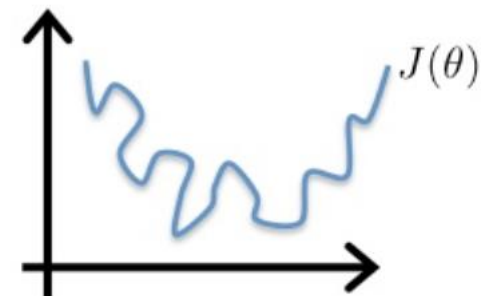
A.



B.



C.

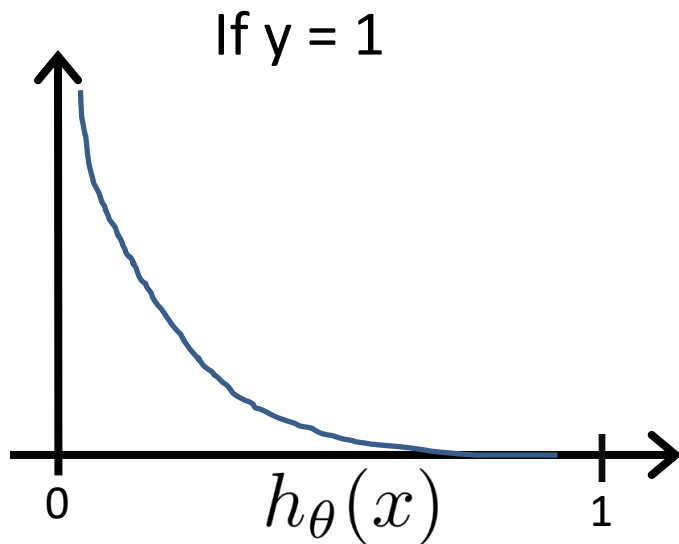


D.

B

Logistic regression cost function

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

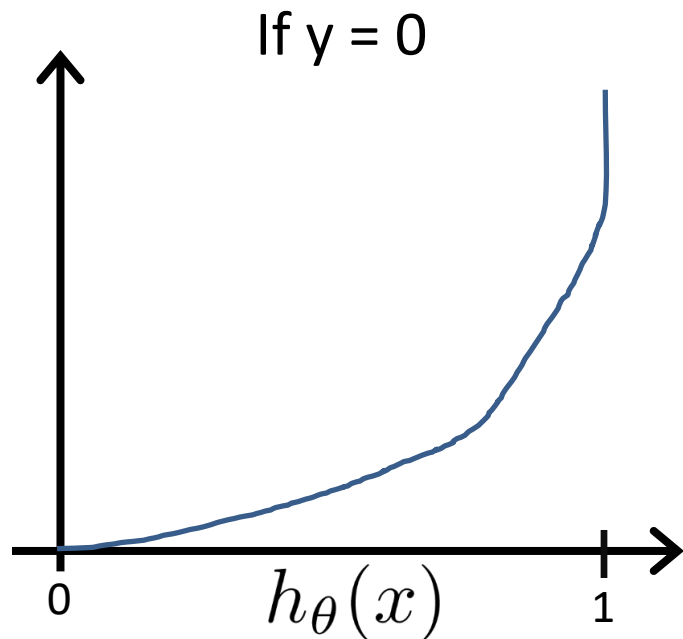


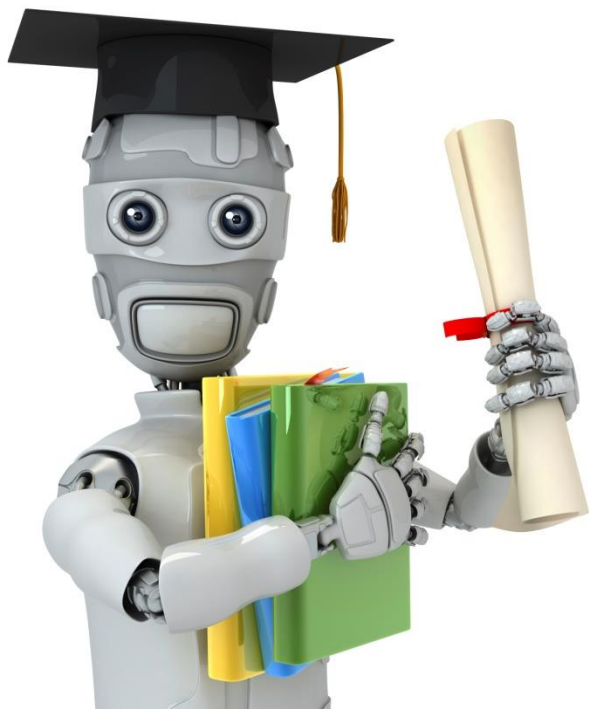
Cost = 0 if $y = 1, h_{\theta}(x) = 1$
But as $h_{\theta}(x) \rightarrow 0$
 $\text{Cost} \rightarrow \infty$

Captures intuition that if $h_{\theta}(x) = 0$, (predict $P(y = 1|x; \theta) = 0$), but $y = 1$, we'll penalize learning algorithm by a very large cost.

Logistic regression cost function

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$





Machine Learning

Logistic Regression

Simplified cost function
and gradient descent

Logistic regression cost function

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)})$$

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

每个变量的代价

Note: $y = 0$ or 1 always

A

$$\text{Cost}(h_{\theta}(x), y) = -y \times \log(h_{\theta}(x)) - (1 - y) \times \log(1 - h_{\theta}(x))$$

$$\text{If } y = 1: \text{Cost}(h_{\theta}(x), y) = -\log(h_{\theta}(x))$$

$$\text{If } y = 0: \text{Cost}(h_{\theta}(x), y) = -\log(1 - h_{\theta}(x))$$

Logistic regression cost function

$$\begin{aligned} J(\theta) &= \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)}) \\ &= -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right] \end{aligned}$$

To fit parameters θ :

$$\min_{\theta} J(\theta)$$

代价最小

To make a prediction given new x :

$$\text{Output } h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

$$P(y = 1|x; \theta)$$

Gradient Descent

$$J(\theta) = -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

Want $\min_{\theta} J(\theta)$:

Repeat {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

}

(simultaneously update all θ_j)

→ 依然保持

同步

Gradient Descent

$$J(\theta) = -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

Want $\min_{\theta} J(\theta)$:

Repeat {

$$\theta_j := \theta_j - \alpha \sum_{i=1}^m (\underline{h_{\theta}(x^{(i)})} - y^{(i)}) x_j^{(i)}$$

}

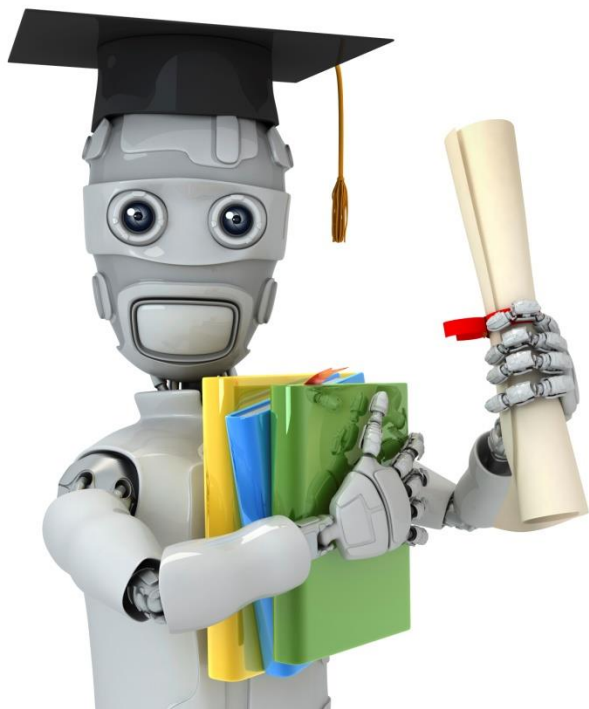
(simultaneously update all θ_j)

→ 同线性

Algorithm looks identical to linear regression!

Suppose that you have trained a logistic regression classifier, and it outputs on a new example x a prediction $h_{\theta}(x) = 0.2$. This means (check all that apply):

- ~~A.~~ Our estimate for $P(y = 1|x; \theta)$ is 0.8.
- B. Our estimate for $P(y = 0|x; \theta)$ is 0.2.
- ~~C.~~ Our estimate for $P(y = 0|x; \theta)$ is 0.8.
- ~~D.~~ Our estimate for $P(y = 1|x; \theta)$ is 0.2.



Machine Learning

Logistic Regression

Multi-class classification:
One-vs-all

Multiclass classification

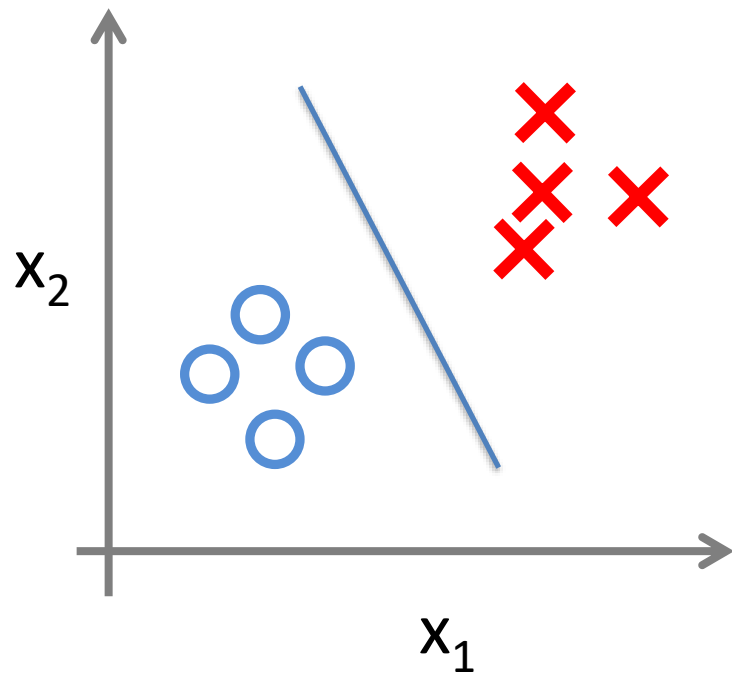
Email foldering/tagging: Work, Friends, Family, Hobby

Medical diagrams: Not ill, Cold, Flu

Weather: Sunny, Cloudy, Rain, Snow

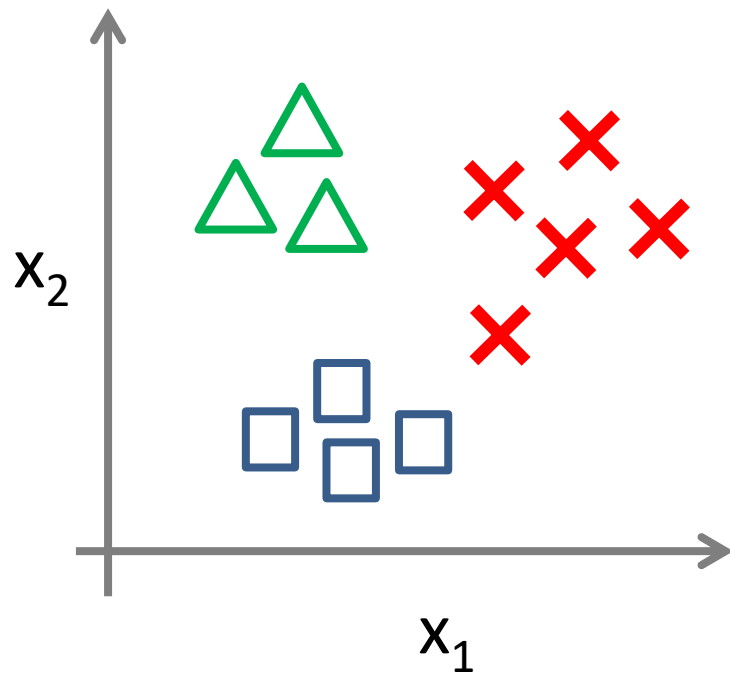
二分

Binary classification:

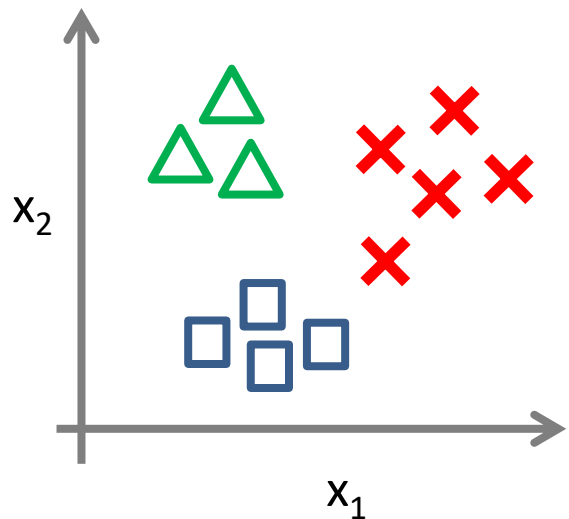


多分

Multi-class classification:



One-vs-all (one-vs-rest):

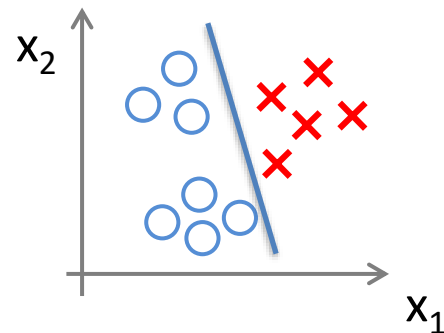
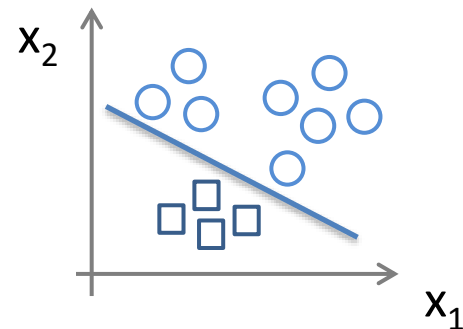
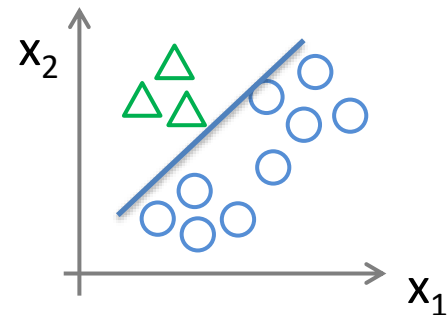


Class 1: 

Class 2: 

Class 3: 

$$h_{\theta}^{(i)}(x) = P(y = i|x; \theta) \quad (i = 1, 2, 3)$$



One-vs-all

Train a logistic regression classifier $h_{\theta}^{(i)}(x)$ for each class i to predict the probability that $y = i$.

On a new input x , to make a prediction, pick the class i that maximizes

$$\max_i h_{\theta}^{(i)}(x)$$

多分类变成每个类的二分类，
然后取出预测的最大值进行分类

是哪一类的概率最大